Routing games

Braess’s paradox, in the room

Phase 1
Two routes. Find where the class settles.

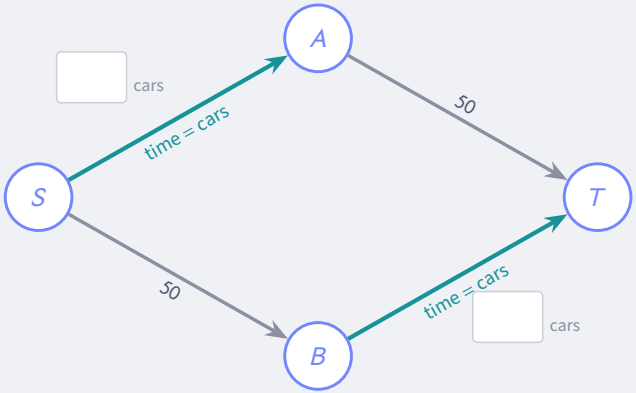
Phase 2
Add a road. Watch it get worse.

Then
Write it down properly.

Drivers in the room:

Date:

Stand on the route you are driving. A small road takes as many minutes as there are people on it; a motorway is fixed at 50 minutes. When the two times are announced you may switch, as often as you like, until nobody gains by moving.



Top route

____ + 50 = ____

Bottom route

____ + 50 = ____

Stand on your route. Re-choose if the other one is quicker. Stop when nobody moves.

Round	On top	On bottom	Top time	Bottom time	Who moves?
1					
2					
3					
4					

Where we settled: ____ on top, ____ on bottom.

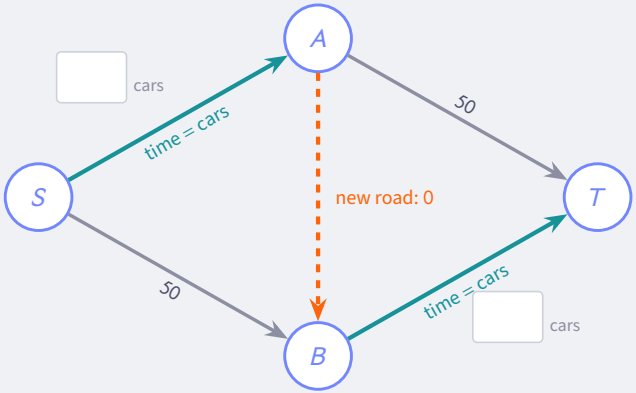
If one driver switches from top to bottom, their time goes from ____ to ____.

So this is a _____ flow.

A flow is a Nash flow when:

Is it also the best the class could do?

Same again, with the new road open. Small roads still take a minute per person and motorways are still 50 minutes. Switch whenever it saves you time, until nobody wants to move.



$S \to A \to T$

$S \to B \to T$

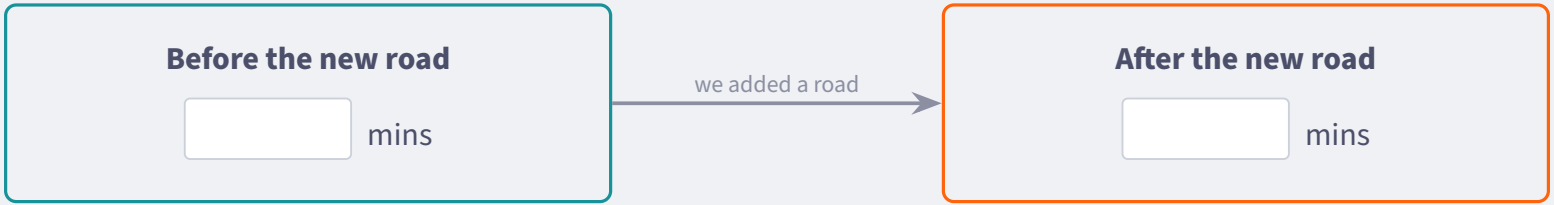
$S \to A \to B \to T$

Re-choose again. The new road is free, so who would not take it?

$S \to A$: ____ cars = ____ mins + $A \to B$: 0 mins + $B \to T$: ____ cars = ____ mins

Everyone's time: ____ mins

Nobody wants to leave: going back to $S \to A \to T$ would cost ____ + 50 = ____.



One more car joins a small road that already carries x cars.

It costs *me* ____ minutes.

It costs *everyone else* ____ minutes.

I pay the first and ignore the second. The road's total time is

$x\,c(x) =$ ____

and one more car really adds $\frac{\mathrm{d}}{\mathrm{d}x}(x\,c(x)) =$ ____

A routing game has a flow f on each route, and a cost

$$C(f) = \sum_e f_e \, c_e(f_e)$$

Our network, *before* the new road:

Our network, *after* the new road:

A planner who could route us all could ignore the new road and send half of us each way.

Left alone: the Nash flow $\tilde{f}$, and each driver's time:

Planned: the optimal flow f^*, and each driver's time:

$$\text{Price of anarchy} = \frac{C(\tilde{f})}{C(f^*)} = \frac{\square}{\square} = \square$$

What we just did in the room is **Question 1** on the Routing Games page, worth 25 marks.

(a)	What a Nash flow is here.	[3]
(b)	The Nash flow and each driver's time.	[5]
(c)	Adding $A \to B$: show all 40 on $S \to A \to B \to T$ is a Nash flow.	[7]
(d)	Compare the times, and explain it through the congestion each driver ignores.	[7]
(e)	Compute the price of anarchy.	[3]

vknight.org/gt/topics/routing-games.html

Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
Choosing a route, over and over	**14.2.1**	Routing Games: Definition: Routing Game
Where nobody gains by switching	**14.2.6**	Routing Games: Definition: Nash Flow
Writing the flows and the costs	**14.2.2 to 14.2.4**	Routing Games: Set of Paths, Feasible Flow, Cost function
What a planner would have done	**14.2.5**	Routing Games: Definition: Optimal Flow
The cost of the extra car	**14.2.9**	Routing Games: Definition: Marginal Cost
Examinable	**14.2.8, 14.2.10**	Nash Flow minimises the potential function, Optimal Flow is a Nash Flow for marginal costs, statements only

vknight.org/gtb/chapters/routing-games

Quick quiz before you go: vknight.org/gt/topics/routing-games.html

Before the next class: read the Routing Games chapter as far as marginal costs, and attempt Question 1.